



Oxytocin inhibits the rat medullary dorsal horn Sp5c/C1 nociceptive transmission through OT but not V_{1A} receptors

Enrique García-Boll, Guadalupe Martínez-Lorenzana, Miguel Condés-Lara, Abimael González-Hernández*

Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus UNAM Juriquilla, Querétaro, QRO, 76230, Mexico

ARTICLE INFO

Article history:

Received 5 August 2017

Received in revised form

16 October 2017

Accepted 17 November 2017

Available online 21 November 2017

Keywords:

Analgesia

Headache

Orofacial

Oxytocin

Pain

Trigeminal

ABSTRACT

The medullary dorsal horn (MDH or Sp5c/C1 region) plays a key role modulating the nociceptive input arriving from craniofacial structures. Some reports suggest that oxytocin could play a role modulating the nociceptive input at the MDH level, but no study has properly tested this hypothesis. Using an electrophysiological and pharmacological approach, the present study aimed to determine the effect of oxytocin on the nociceptive signaling in the MDH and the receptor involved. In sevoflurane, anesthetized rats, we performed electrophysiological unitary recordings of second order neurons at the MDH region responding to peripheral nociceptive-evoked responses of the first branch (V1; ophthalmic) of the trigeminal nerve. Under this condition, we constructed dose-response curves analyzing the effect of local spinal oxytocin (0.2–20 nmol) on MDH nociceptive neuronal firing. Furthermore, we tested the role of oxytocin receptors (OTR) or vasopressin V_{1A} receptors (V_{1A}R) involved in the oxytocin effects. Oxytocin dose-dependently inhibits the peripheral-evoked activity in nociceptive MDH neurotransmission. This inhibition is associated with a blockade of neuronal activity of A δ - and C-fibers. Since this antinociception was abolished by pretreatment (in the MDH) with the potent and selective OTR antagonist (L-368,899; 20 nmol) and remained unaffected after the V_{1A}R antagonist (SR49059; 20 nmol or 200 nmol), the role of OTR is implied. This electrophysiological study demonstrates that oxytocin inhibits the peripheral-evoked neuronal activity at MDH, through OTR activation. Thus, OTR may represent a new potential drug target to treat craniofacial nociceptive dysfunction in the MDH.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

The medullary dorsal horn (MDH) is involved in pain processing from the craniofacial area (Hu et al., 2005; Mørch et al., 2007; Tsai et al., 1999) and modulates nociceptive information arriving from the first branch of the trigeminal nerve (V1, ophthalmic branch) (Meng et al., 1997; Malick et al., 2000). This branch nerve innervates the periorbital area, intranasal mucosa and supratentorial dural and pial tissues (Shankland, 2001). Noxious stimuli carried by this branch through A δ - and C-fibers arrive at the MDH where they contact second order neurons projecting to supraspinal sites (Hu, 1990; Shimizu and Suzuki, 2011). At this level, the nociceptive information can be endogenously or exogenously modulated by an

array of neuromediators (Akerman et al., 2011; Goadsby et al., 2009). Indeed, some drugs are therapeutically relevant; for example, the 5-HT_{1B/1D} receptor agonists (triptans) and CGRP receptor antagonists (gepants), both of which are specific antimigraine compounds that selectively inhibit the nociceptive input arriving at MDH or Sp5c/C1 region (Kaube et al., 1993; Storer et al., 2004).

A research in humans with tension-type headache or migraine has suggested that intranasal oxytocin seems to have analgesic effects (Wang et al., 2013); regardless, the locus of action and receptors involved remain unknown. A potential role of oxytocin at MDH level is suggested considering that after intranasal administration of ¹²⁵I-labeled oxytocin in rats, several areas involved in pain transmission, including Sp5c are marked (Tzabazis et al., 2017). Although this neuropeptide has emerged as an interesting molecule with analgesic properties when given spinally (at lumbar level in humans) (Condés-Lara et al., 2016), the receptors involved

* Corresponding author.

E-mail address: abimaelgh@comunidad.unam.mx (A. González-Hernández).

URL: http://132.248.142.23/web_site/home_pages/129

remain unclear (González-Hernández et al., 2014). Hence, the potential antinociceptive actions of oxytocin and receptors implicated in MDH neurotransmission deserves attention.

Considering that oxytocin binds to both OT receptors (OTR; pK_i : 8.2) and vasopressin V_{1A} receptors ($V_{1A}R$; pK_i : 8.3) with similar affinities (Alexander et al., 2015), the present study was designed using an electrophysiological approach to elucidate the neuropharmacological basis of the following: (i) the effect of oxytocin on Sp5c/C1 nociceptive responses evoked by electrical stimulation of the supraorbital dermatome (which is mostly innervated by the V1 branch); and (ii) the role of OTR or $V_{1A}R$ in the oxytocin-induced inhibition by analyzing the effects of the selective and potent antagonists L-368,899 (OTR; pK_i : 8.1, Williams et al., 1994) or SR-49059 ($V_{1A}R$; pK_i : 8.1–9.3, Serradeil-Le Gal et al., 1993). Our results show that at MDH, oxytocin inhibits the peripheral nociceptive input through OTR but not through $V_{1A}R$.

2. Material and methods

2.1. Animals

A total of 56 adult male Wistar rats (260–320 g) provided by our institutional animalarium were used. The animals were housed for at least four days in plastic cages (1 rat per cage) with on wood-based bedding, controlled temperature ($23 \pm 2^\circ\text{C}$) and lighting (12/12-h light-dark cycle; light beginning at 7 a.m.), with access to food and water *ad libitum*. Our Institutional Ethics Committee approved all animal procedures and we followed the regulations established by the Official Mexican Standard (NOM-062-ZOO-1999), which fully complies with (i) the ARRIVE guidelines and (ii) the guide for the Care and Use of Laboratory Animals established by the National Institutes of Health (NIH Publications No. 8023, revised 1978).

2.2. General methods

2.2.1. Surgical procedures

Animals were introduced into a chamber and deeply anesthetized with 6% (v/v) sevoflurane delivered through a vaporizer at 3:1 ratio of N_2O and O_2 . The adequacy of anesthesia before surgery was judged by the absence of ocular reflexes, a negative tail flick test and corporal relaxation. Under these conditions, an intratracheal cannula was inserted for artificial ventilation (56 strokes/min) and to maintain the anesthesia throughout the experiments. The stroke volume was adjusted to maintain a normal acid-base equilibrium. Rats were mounted on a stereotaxic frame (Kopf Instruments, Tujunga, CA). The muscles of the dorsal neck were separated to access the MDH. A cervical (C1) laminectomy was performed and the dura was carefully removed to expose the brainstem, at the level of the caudal medulla oblongata. The subsequent protocols were performed under 2.0% sevoflurane to achieve the ethically adequate level of anesthesia without depressing neuronal responses to noxious stimuli. The animals were paralyzed with a single pancuronium bromide injection (2 mg/kg; intramuscular). End-tidal CO_2 was monitored (Capstar-100, CWI Inc., Ardmore, PA, USA), and core body temperature was maintained at 38°C using a circulating water pad. At the end of each experiment, animals were perfused transcardially (see Section 2.3).

2.2.2. Extracellular unitary recordings at medullary dorsal horn

Extracellular unitary recordings of MDH neurons with input from the supraorbital receptive field were made using quartz-insulated platinum/tungsten microelectrodes (4–9 M Ω) mounted in a microdrive Eckhorn system (Thomas Recording GmbH, Giessen, Germany). This device allows the insertion of 3 microelectrodes (distance inter-electrode: 300 μm) into neural tissue and

each microelectrode was advanced into the MDH independently using the Eckhorn Matrix multiuser software (Thomas Recording GmbH, Giessen, Germany). In most cases we recorded only one extracellular unit per animal and only in four animals we recorded two extracellular signals (see 3.1 section). To search for single-unit discharges, the microelectrodes were lowered (400–1600 μm from the surface) in small steps (2–5 μm). A first search of cells responding to tactile stimulation of the ophthalmic dermatome was performed. In all cases, the specific receptive field was exclusively assessed in the first division of the trigeminal nerve (V1; ophthalmic). The receptive field was assessed for non-noxious (brushing) and noxious (pinching) input; then, electrical stimulation was applied by two electrodes inserted into the receptive field. In this case, two fine needles (27 G) attached to a stimulus isolator unit were inserted subcutaneously into the receptive field of the recorded neuron. Then, electrical test stimulation was conducted. This test consists of 20 stimuli at 0.5 Hz with 1-msec pulse duration at 1.5 times the threshold intensity (0.1–3 mA) required to evoke a A δ - and C-fiber response.

The evoked extracellular neuronal activity was recorded, amplified, digitalized, and discriminated using CED hardware and Spike2 v5.15 software (Cambridge Electronic Design). Raw and discriminated signals were fed through an audio monitor and displayed on an oscilloscope. Waveforms were stored on a hard drive for off-line analysis. Baselines and evoked activities of the MDH wide dynamic range (WDR) neurons were recorded and analyzed as post-stimulus time histograms (PSTH) to detect the occurrence of statistically significant neuronal responses. On this basis, the stimulating threshold to evoke action potentials and their frequency of occurrence, resulting from the stimulation of the peripheral receptive field located on the supraorbital area, were attributed to the recruitment of A δ - and C-fibers. Considering the distance between the receptive field and the recording electrode, the peak latencies observed correspond to peripheral conduction velocities within the A δ - (3–25 msec) and C-fibers (25–80 msec) (see Fig. 1 for details). Thus, the number of action potentials that occurred in response to 20 receptive field stimuli was compared before (basal) and after vehicle/drug treatment.

The evoked neuronal responses were evaluated after oxytocin (0.2, 2 and 20 nmol; $n = 6$ cells, each concentration) or vehicle ($n = 6$ cells) administration at 10, 20, 30, 40, 50 and 60 min. The role of OTR or $V_{1A}R$ in the oxytocin effect was also evaluated using the competitive antagonists L-368,899 (20 nmol; $n = 6$ cells) or SR-49059 (20 or 200 nmol; $n = 6$ cells each group) (Pettibone et al., 1993; Serradeil-Le Gal et al., 1993). The antagonist was administered 5 min before oxytocin or vehicle. All drugs/vehicles were carefully delivered at spinal level (topical) where the microelectrodes were positioned in a volume of 20 μl using a Hamilton[®] syringe.

Finally, it was of the foremost importance to ascertain if MDH administration of OTR ($n = 6$ cells), $V_{1A}R$ ($n = 6$ cells) antagonist or its vehicle ($n = 6$ cells) was devoid of effect *per se* on the nociceptive-evoked responses; accordingly, any effect of a given antagonist on the oxytocin-induced inhibition is due to a direct interaction of the antagonist with its respective receptor.

2.3. Histological reconstruction of recording sites

At the end of the electrophysiological experiments, the animals received an overdose of sevoflurane (8% v/v) and were transcardially perfused with saline solution followed by 10% formaldehyde (~200 ml of each). Brains were post-fixed in 10% formaldehyde and cut into 40- μm serial coronal sections with a freezing microtome (Leica Instruments BmbH, Nussloch, Germany). The position of the cells was observed with a light microscope.

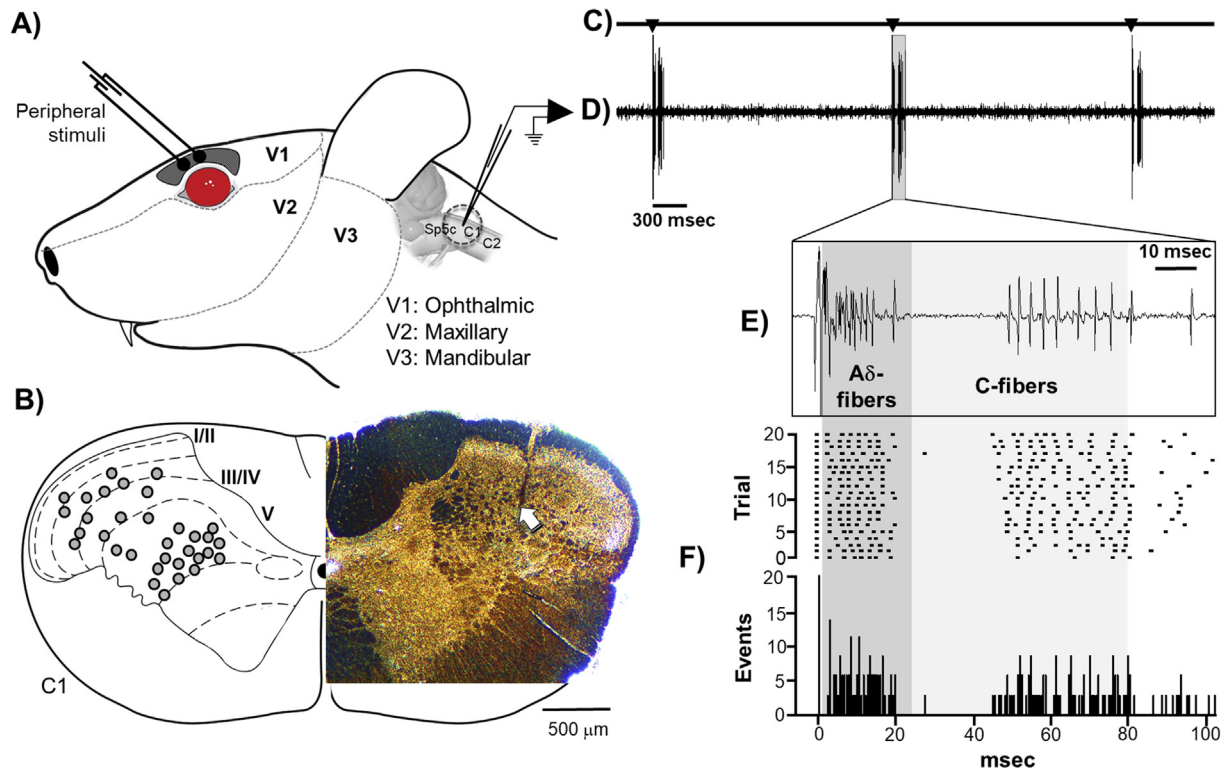


Fig. 1. Experimental set-up used to record second order wide dynamic range (WDR) neurons at MDH. (A) Diagram showing the location of the cutaneous receptive field (in grey) of all neurons studied. This receptive field correspond to the first division of the trigeminal nerve (V1; ophthalmic branch). All neurons were recorded using a quartz platinum/tungsten microelectrodes (4–9 MΩ) inserted into the Sp5c/C1 region. (B) Summary of the locations of recording sites (left) in the Sp5c/C1 region and example (right; microphotography) of a lesioned recording site in the dorsal horn at C1 level. Panel (C) shows three stimulus artifacts induced by receptive field electrical stimulation and (D) the three neuronal responses (on a second order wide dynamic neuron) induced by the electrical stimuli. (E) Raw tracing of a single typical response induced by one electrical stimulus on the receptive field depicting the time latencies of Aδ-fiber and C-fiber components. (F) A raster plot (above) and a peri-stimulus time histogram (below) obtained from 20 responses to 20 electrical stimuli; the time latencies depict the fiber components (Aδ-fibers, 5–25 msec; and C-fibers, 25–80 msec).

2.4. Drugs

Apart from the anesthetics, this study used the following compounds (obtained from the sources indicated): oxytocin acetate salt hydrate (Sigma Chemical Co., St Louis, MO, USA); (2S)-2-amino-N-[(1S,2S,4R)-7,7-dimethyl-1-[[[(4-(2-methylphenyl)-piperazinyl)sulfonyl] methyl]bicyclo[2.2.1]hept-2-yl]-4-(methylsulfonyl)butanamide hydrochloride (L-368,899 hydrochloride; Tocris Bioscience, Northpoint, UK); and (2S)-1-[[[(2R,3S)-5-Chloro-3-(2-chlorophenyl)-1-[(3,4-dimethoxyphenyl)sulfonyl]-2,3-dihydro-3-hydroxy-1H-indol-2-yl]carbonyl]-2-pyrrolidinecarboxamide (SR-49059 or relcovaptan; Tocris Bioscience, Northpoint, UK). Oxytocin was dissolved in physiological saline. L-368,899 and SR-49059 was dissolved in 10% DMSO and the resulting solution was gauged with saline.

2.5. Data presentation and statistical evaluation

The temporal course figures in the text are presented as mean $\pm$ standard error of the mean (S.E.M.), and we used the box-and-whiskers plots, to compare the global neuronal effect of oxytocin treatments. In all cases, the number of evoked potentials induced by electrical stimulation of the cutaneous receptive fields in the ophthalmic division (V1) of the trigeminal nerve was normalized and expressed as percentage change from the respective baseline. This baseline response was established after an identified neuron had a $\leq 10\%$ variation in the neuronal responses induced by receptive field stimulation during 5 consecutive tests

(5 min between each test) and no outliers was identified (using the Grubbs' test). To evaluate only the stability of the recorded neurons (control group and vehicle group) across the 60-min time frame, we used one-way analysis of variance (ANOVA) for repeated measures; sphericity was not assumed and corrections to degrees of freedom were made according to Greenhouse-Geisser method. Furthermore, to compare the values of neuronal Aδ- or C-fiber responses obtained before (basal response) and after drug treatments in the different subgroups, we used a two-way ANOVA for repeated measures followed by the Student-Newman-Keuls (SNK) test (if applicable) for multiple comparisons. Finally, to compare the global neuronal activity between different treatments we used a one-way ANOVA followed by a Bonferroni *post-hoc* test (if applicable). In all cases, $P \leq 0.05$ was considered statistically significant.

3. Results

3.1. Electrophysiological data

Recordings were made from a total of 60 neurons in 56 animals considering that in four rats we recorded simultaneously two WDR neurons. Animals studied had a mean body weight of 290 ± 30 g. Extracellular recording in the MDH were made from WDR neurons with cutaneous receptive fields (supraorbital) in the ophthalmic division (V1) of the trigeminal nerve, with a similar distribution to the one previously reported (Hu et al., 2005; Mørch et al., 2007). Most neurons (~80%) were in lamina V of the dorsal horn at the level of the cervicomedullary junction (with a few neurons in

laminae II–IV <20%), at an average depth of 1145 μm . These neurons were characterized according to the properties of their responses to peripheral receptive field stimulation and were classified as WDR neurons since they were activated by non-noxious and noxious stimuli and presented A δ -, and C-fiber responses after peripheral receptive field electrical stimulation (Fig. 1).

3.2. Stability of electrophysiological recordings

Since we recorded MDH nociceptive neurons during 60-min, we first evaluated the effect of time on the neuronal responses. In the control group ($n = 6$), time had no effect on A δ -fibers ($F_{1,1994, 9.968} = 2.190, P = 0.1629$) or C-fibers ($F_{2,130, 10.65} = 0.9711, P = 0.4151$) responses of trigeminal second order neurons during the time recorded. Also, vehicle administration had no effect on A δ -fiber ($F_{2,366, 11.83} = 0.6938, P = 0.5418$) or C-fiber ($F_{2,689, 13.44} = 2.902, P = 0.0780$) responses of these neurons ($n = 6$) across the time. Together, these data show that no time-dependent changes in neuronal responses occurred during our experimental protocols.

3.3. Spinal administration of oxytocin dose-dependently inhibits the peripheral nociceptive-evoked activity of the second order WDR neuron in the MDH region

Fig. 2 shows the A δ - and C-fiber responses induced by electrical receptive field stimulation in control cells ($n = 6$) and cells pretreated with oxytocin (0.2, 2 and 20 nmol; $n = 6$ cells per concentration). In the case of A δ -fibers (Fig. 2A), the two-way ANOVA for repeated measures revealed that oxytocin ($F_{3, 20} = 4.450, P = 0.015$) affects the peripheral induced-neuronal activity of trigeminal second order neurons in a time-dependent manner ($F_{6, 120} = 7.367, P < 0.0001$); indeed, interaction was detected ($F_{18, 120} = 2.181, P = 0.0068$). Using the pairwise multiple comparison procedure (SNK *post hoc* test) we found that this inhibition was particularly patent (~20–25%) for 20 nmol oxytocin, starting at 20 min and lasting up to 60 min. Similarly, 2 nmol oxytocin inhibits MDH neuronal activity (~20%), but this inhibition was observed at 50 and 60 min. When we analyzed these results as global neuronal firing (Fig. 2B) using the one-way ANOVA ($F_{3, 140} = 24.12, P < 0.0001$) followed by the Bonferroni *post hoc* test, we found that oxytocin blocked the peripheral nociceptive A δ -fiber input in a dose-dependent manner.

Similar results were obtained in the case of C-fibers (Fig. 2C). The statistical analysis revealed that treatments ($F_{3, 20} = 8.469, P = 0.0008$) affect the peripherally induced-neuronal activity of trigeminal second order neurons and depend on time ($F_{6, 120} = 11.58, P < 0.0001$); interaction also was detected ($F_{18, 120} = 6.686, P < 0.0001$). Using the SNK *post hoc* test, we found that this inhibition was observed for: (i) 2 nmol oxytocin (~25–50%; starting at 20 min) and (ii) 20 nmol oxytocin (~45–70%), starting at 10 min. When we analyzed these results as global neuronal firing (Fig. 2D) using the one-way ANOVA ($F_{3, 140} = 43.71, P < 0.0001$) followed by the Bonferroni *post hoc* test, we found that oxytocin inhibited the peripheral nociceptive C-fiber input in a dose-dependent manner.

Fig. 2E exemplifies the oxytocin-induced inhibition of the neuronal nociceptive activity of the MDH neurotransmission, and it shows a single unit evoked neuronal response (of the same neuron) before and after (min 50) oxytocin (20 nmol) administration. Note that PSTH constructed for this neuron (Fig. 2F) shown that the neuronal events induced by 20 electrical pulses at peripheral level diminished after oxytocin; an effect particularly patent for the C-fiber component. Taken together, these data show that oxytocin inhibit the nociceptive input of peripherally-evoked trigeminocervical activation.

3.4. Oxytocin receptor but not vasopressin V_{1A} receptor antagonist abolishes the oxytocin-induced antinociception in the MDH

To investigate the receptor involved in the oxytocin-induced antinociception, a selective antagonist to OTR or $V_{1A}R$ was used. Fig. 3 shows the effect of pretreatment with L-368,899 (20 nmol; OTR antagonist) on the neuronal MDH responses of A δ - and C-fibers treated with oxytocin (20 nmol). In the case of A δ -fibers pretreated with L-368,899 or vehicle (Fig. 3A), the two-way ANOVA for repeated measures revealed that neither treatments ($F_{2, 15} = 0.6611, P = 0.5307$) nor time ($F_{6, 90} = 0.7784, P = 0.5890$) affect the peripherally induced neuronal activity of trigeminal second order neurons, suggesting that pretreatment with this antagonist blocked the oxytocin-induced antinociception. Similar results were obtained in the case of C-fibers (Fig. 3B); $F_{2, 15} = 0.8131, P = 0.4621$ for treatments and $F_{6, 90} = 0.7215, P = 0.6333$ for time. In both cases (Fig. 3, insets), the one-way ANOVA did not found any difference in the global neuronal firing ($F_{2, 105} = 2.73, P = 0.0671$, A δ -fibers activity; $F_{2, 105} = 4.338, P = 0.0655$, C-fibers activity).

In contrast, as Fig. 4 shown the two-way ANOVA for repeated measures revealed that pretreatment with SR-49059 (20 nmol or 200 nmol) has no effect on the oxytocin-induced inhibition of A δ - and C-fibers activity. Indeed, in the case of A δ -fibers (Fig. 4A), the two-way ANOVA for repeated measures comparing the treatment groups demonstrated the main effect as differences between groups ($F_{3, 20} = 6.686, P = 0.0026$) and interactions ($F_{18, 120} = 2.605, P = 0.0011$). The SNK *post hoc* test showed that only SR-49059 plus oxytocin inhibited neuronal activity of A δ -fibers in the MDH (~15–25%), starting at minute 10 and lasting up to minute 60. When we analyzed these results as global neuronal firing (Fig. 4B) using the one-way ANOVA ($F_{3, 140} = 34.65, P < 0.0001$) followed by the Bonferroni *post hoc* test, we found that 20 or 200 nmol SR-49059 did not have any effect on the oxytocin-induced inhibition of A δ -fibers activity.

Similar effects were observed in the case of C-fibers (Fig. 4C): $F_{3, 20} = 9.939, P = 0.0003$ for treatments; $F_{6, 120} = 13.64, P < 0.0001$ for time; $F_{18, 120} = 4.560, P < 0.0001$ for interactions. The SNK *post hoc* test showed that only SR-49059 plus oxytocin inhibited the neuronal responses of C-fibers (~45–60%), starting at minute 10 and lasting up to minute 60. Analyzing these results as global neuronal firing (Fig. 4D) and using the one-way ANOVA ($F_{3, 140} = 44.41, P < 0.0001$) followed by the Bonferroni *post hoc* test, we found that 20 or 200 nmol SR-49059 did not have any effect on the oxytocin-induced inhibition of C-fibers activity.

4. Discussion

4.1. General

This electrophysiological study aimed to understand the neuropharmacological basis of the potential role of oxytocinergic mechanisms modulating nociceptive neurotransmission in second order neurons at MDH region. Regarding our experimental protocols, we did not measure the activity of the primary afferent fibers (PAF) A δ - or C-fibers directly; instead, we analyzed a particular set of second order neurons, namely WDR cells that respond to peripheral electrical stimuli delivered at the supraorbital area (mostly innervated by the first branch of trigeminal nerve). Apart from the implications discussed below, the data demonstrate that oxytocin can modulate nociceptive MDH neurons. We show that when given oxytocin at this level, the number of action potentials evoked peripherally and related to of A δ - and C-fibers activation are drastically suppressed (>50% for C-fibers and ~25% for A δ -fibers; Fig. 2). Furthermore, since the oxytocin-induced inhibition of nociceptive transmission at MDH was abolished by pretreatment

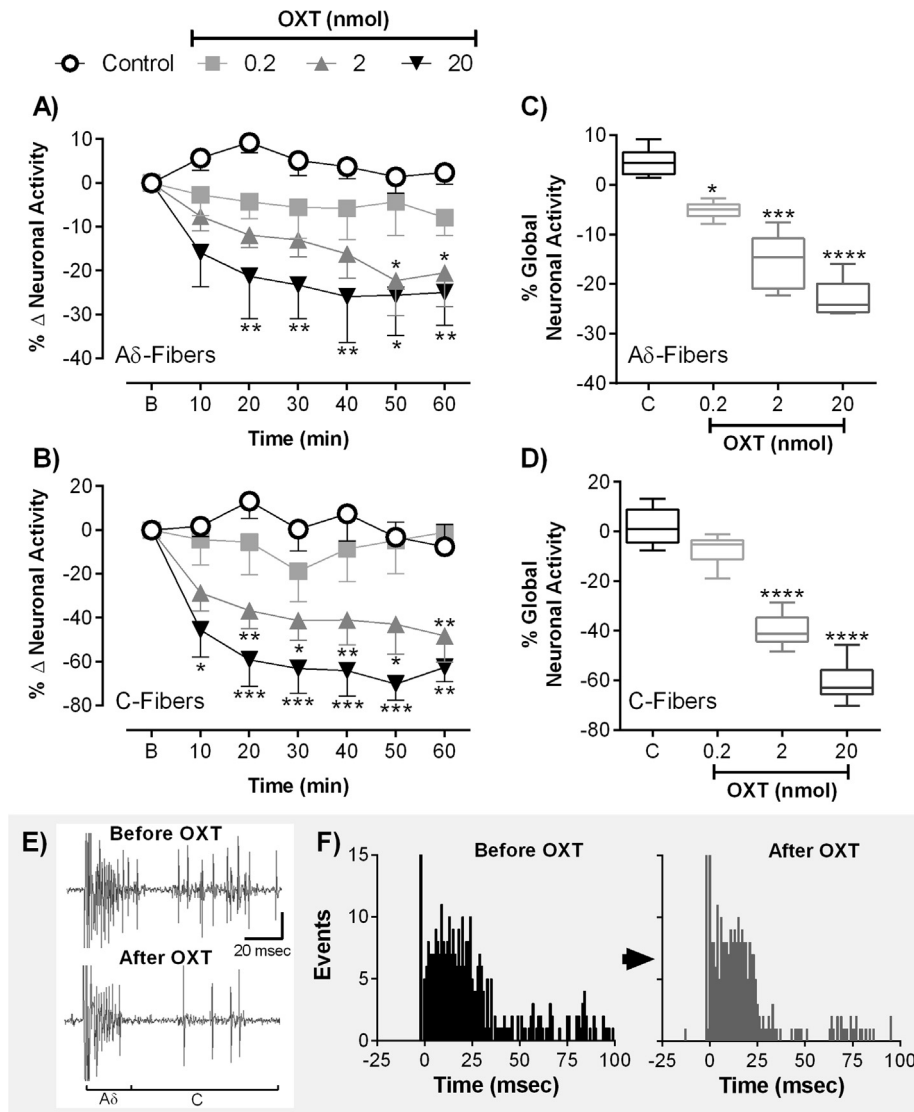


Fig. 2. Spinal oxytocin inhibits the nociceptive peripheral evoked neuronal firing in the second order wide dynamic range (WDR) MDH neurons. (A and C) Time course changes in the percentage average of the Aδ- and C-fiber neuronal firing induced by the receptive field electrical stimulation and the effects to topical spinal oxytocin administration (0.2, 2 and 20 nmol; $n = 6$ cells each concentration). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ statistically significant difference compared at the same time with control response ($n = 6$ cells). Note that 20 nmol oxytocin inhibits both components of the neuronal activity (i.e. Aδ- and C-fiber activation). (C and D) These panels show the global neuronal activity (obtained from the respective time courses) of Aδ- and C-fibers in response to spinal oxytocin. * $P < 0.05$, *** $P < 0.001$, **** $P < 0.0001$ vs. control. (E) Show the unitary response (raw data) of the same second order neuron in response to one electrical stimulation before (basal response) and after (40 min) oxytocin administration (20 nmol). Note that the number of spikes after oxytocin are clearly diminished. (F) Peri-stimulus time histogram obtained from 20 responses to 20 electrical stimuli before (basal) and after (40 min) 20 nmol oxytocin administration; note that, the number of events diminished after oxytocin, particularly for the C-fiber component.

with an OTR antagonist and remained unaffected after the $V_{1A}R$ antagonist, a specific role of OTR modulating the nociceptive input is suggested (Figs. 3, 4 and 5). These results agree with the antinociceptive role of oxytocin in other animal models of different pain states. Furthermore, to our knowledge, this is the first study that directly investigates the role of oxytocinergic mechanisms modulating neurotransmission at MDH region, a key area involved in craniofacial nociceptive transmission.

4.2. Oxytocin blocks the nociceptive input to the MDH neurons through OTR

Since oxytocin inhibits the MDH nociceptive trigeminal neurotransmission (Fig. 2), oxytocinergic mechanisms are implied. Previous electrophysiological reports at spinal lumbar level showed

that 10^{-6} M oxytocin (i.e. 0.2 nmol in 20 μ l) inhibits the nociceptive activity of WDR neurons (Rojas-Piloni et al., 2007). In our experiments 0.2 nmol has not any effect on the evoked neuronal nociceptive transmission, an effect probably related with the fact that the transcriptome expression protein profile in the dorsal root ganglia (innervating the spinal cord) and trigeminal ganglia (innervating MDH) seems to be different (Kogelman et al., 2017). Moreover, although 2 nmol decrease the global neuronal activity of Aδ- and C-fibers (Fig. 2C and D), only 20 nmol induced a consistent antinociception along the time (Fig. 2A and B).

In the case of 2 nmol oxytocin, the fact that this molecule inhibited the Aδ-fiber activity only at late time points could be partially explained considering pharmacodynamic and pharmacokinetic factors, namely: (i) the low oxytocin concentrations have low probability to interact with its receptor and consequently

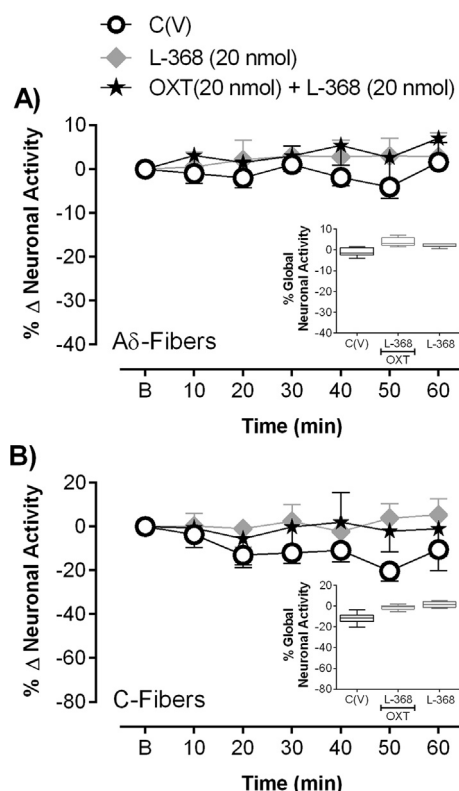


Fig. 3. Role of OTR in the spinal oxytocin-induced antinociception in the MDH. Panels (A) and (B) show the effect of pretreatment with the OTR antagonist (20 nmol L-368,899) on the oxytocin-induced antinociception on second order Sp5c/C1 neurons; this antagonist abolishes the oxytocin-induced inhibition of neuronal activity associated with Aδ- and C-fiber activation.

induce a response; in addition (ii) the time necessary to reach the site of action located at an average depth of 1145 μm from the surface. Furthermore, although we do not have an explanation for the long-lasting inhibition of nociceptive integration in the MDH by oxytocin, has been reported that pharmacological (González-Hernández et al., 2017) or optogenetic (Eliava et al., 2016) activation of oxytocinergic fibers elicited a long-lasting inhibition (>40 min) of C-fibers discharge recorded in WDR neurons.

In physiological terms, it has been shown that a direct oxytocinergic projection from the paraventricular nuclei (PVN, a main source of oxytocin) arrives at the spinal dorsal horn (Swanson and McKellar, 1979) and upon stimulation induces antinociception (Condés-Lara et al., 2006). Likewise, a dense PVN projection to the trigeminal nucleus caudalis (Sp5c), where ophthalmic afferents project has been demonstrated (Abdallah et al., 2013; Robert et al., 2013), supporting our contention about the potential role of oxytocinergic mechanisms modulating MDH nociceptive neuronal activity. Furthermore, a recent report in rats (Tzabazis et al., 2017) showed that intranasal (an area innervated by the V1 and V2 branch of trigeminal nerve) ^{125}I -labeled oxytocin not only diffuses to the Sp5c and several structures implicated in pain processing but also diminishes Sp5c neuronal activity. Hence, the role of oxytocin modulating Sp5c/C1 neurotransmission is suggested, but we cannot exclude that due to the type of administration (intranasal), several central structures are recruited and could globally exert a descending modulatory action of MDH nociceptive activity. Indeed, supraspinal oxytocin could block pain transmission by enhancing descending analgesics mechanisms (Caldwell et al., 1987; Condés-Lara et al., 2015; Ge et al., 2002; Lundberg et al., 1994; Wang et al., 2003).

Our data provide direct evidence about the role of oxytocinergic mechanisms at Sp5c/C1 level and introduce the idea about the potential role of OTR. However, since we used a pharmacological strategy oxytocin could bind not only OTR but also $V_{1A}R$, and an unspecific effect on this receptor could be contributing to the oxytocin effect. Certainly, the main receptor involved (OTR vs. $V_{1A}R$) in the oxytocin-induced antinociception at spinal dorsal horn is a matter of debate (González-Hernández et al., 2014).

To identify the pharmacological profile of the receptors involved we used the OTR antagonist, L-368,899 (rat pK_i : 8.1; 50-fold selectivity over $V_{1A}R$ receptors; Williams et al., 1994; Thompson et al., 1997) and the $V_{1A}R$ antagonist, SR-49059 (rat pK_i : 8.1–9.3; >50-fold selectivity over OTR; Serradeil-Le Gal et al., 1993). Although several pharmacological studies have been using these antagonists to discriminate between OTR and $V_{1A}R$ (Gupta et al., 2008; Mack et al., 2007; Thompson et al., 1997; Wilson et al., 2009) the concentration necessary to block OTR or $V_{1A}R$ at MDH has not been reported. In this sense, we decided to use an equimolar concentration (i.e. the same concentration of oxytocin needed to block nociceptive transmission, 20 nmol) considering that oxytocin exhibits a similar affinity for the OTR (pK_i : 8.2) and $V_{1A}R$ (pK_i : 8.3) than the antagonists used (Alexander et al., 2015).

As shown in Figs. 3, 4 and 5, the fact that oxytocin-induced inhibition was abolished by L-368,899 and remained unaffected after SR-49059 supports the role of OTR. This finding is strongly reinforced considering that a supramaximal dose of SR-49059 (200 nmol) did not have any effect on the oxytocin-induced inhibition. Hence, our findings reveal that the oxytocin-induced antinociception at MDH is OTR dependent.

4.3. Further evidence supporting the role of OTR at MDH level

Our findings showing that oxytocin-induced inhibition of the nociceptive responses is mediated by OTR activation in the MDH region are consistent with a pair of studies measuring OTR in trigeminal ganglia neurons (Kubo et al., 2017; Tzabazis et al., 2016). In this respect, it has been shown that: (i) OTR co-localizes with CGRPergic nociceptive fibers; (ii) oxytocin blocks the capsaicin-induced CGRP release from dural nociceptors (using an *ex-vivo* model); and (iii) under orofacial inflammation OTR expression increases. Nevertheless, $V_{1A}R$ is also expressed in trigeminal ganglia (Kubo et al., 2017), but considering our pharmacological data it is reasonable to assume that this receptor is not involved in the oxytocin effect. Certainly, additional studies that fall beyond the scope of this investigation will be required to ascertain the role of $V_{1A}R$ in the nociception at MDH. Additional indirect evidence pointing out the relevance of OTR modulating nociception comes from studies using mice. These studies show that oxytocin-binding sites are restricted to the superficial layers of the dorsal horn, whereas vasopressin binding sites are slightly and diffusely present in the spinal cord gray matter (Schorscher-Petcu et al., 2010). In any case, these studies support our contention that OTR probably present in the PAFs reaching the MDH are involved in the oxytocin-induced craniofacial antinociception.

4.4. Possible locus and mechanisms by which oxytocin administration result in a MDH inhibition of nociceptive transmission

Certainly, additional studies will be required to ascertain the locus of OTR mediating inhibition of craniofacial nociception and how this receptor activation can selectively block the nociceptive transmission, but we could delineate two probably routes:

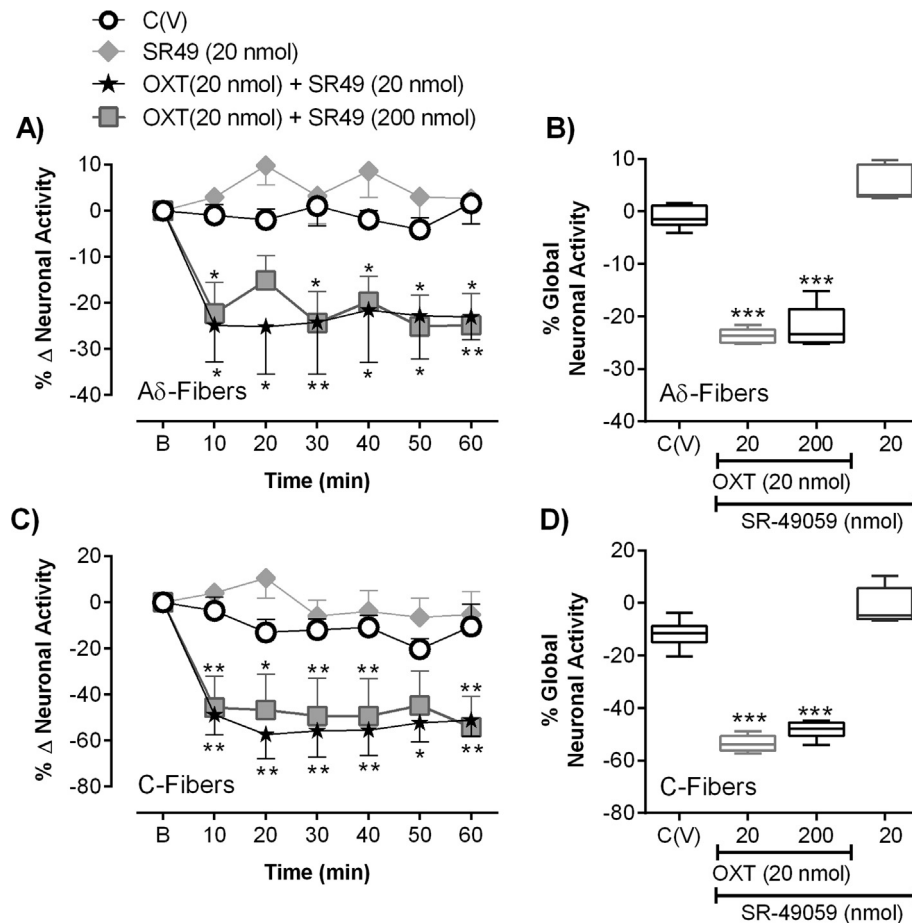


Fig. 4. Exclusion of $V_{1A}R$ in the spinal oxytocin-induced antinociception in the MDH. Panels (C) and (D) shows that pretreatment with the $V_{1A}R$ antagonist (20 nmol or 200 nmol SR-49059) had no effect on the oxytocin-induced antinociception. Panels (C) and (D) illustrates the global neuronal activity (obtained from the respective time courses) of Aδ- and C-fibers between the different treatments; note that SR-49059 pretreatment did not have any effect (even at supramaximal dose, 200 nmol) on the oxytocin-induced inhibition of nociceptive responses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ vs C(V) (control group with vehicle).

- (i) First, a direct inhibition of the C-fibers discharge considering that OTR is expressed in these nociceptive PAFs of the trigeminal ganglia neurons (Kubo et al., 2017; Tzabazis et al., 2016). Indeed, in sensory nociceptive fibers from the dorsal root ganglia it has been shown that OTR activation induced a large inhibition of KCl depolarization-induced increase in intracellular Ca^{2+} concentration (Hobo et al., 2012), and consequently favored the membrane hyperpolarization by activation of the intracellular nNOS/NO/ K_{ATP} pathway (Gong et al., 2015), a mechanism associated with antinociception (Brito et al., 2006).
- (ii) Second, considering that PVN oxytocinergic projections could be reaching on inhibitory interneurons innervating the primary sensory neurons, we could suggest an indirect action of oxytocin. Electrophysiological studies showed that the neuronal activity of Aδ- and C-fibers could be inhibited indirectly by OTR activation on glutamatergic neurons which, in turn, amplify the activity of GABAergic interneurons innervating the PAFs (Breton et al., 2008; Rojas-Piloni et al., 2007). Furthermore, at this level opioid- or serotonergic-dependent mechanisms have been also proposed (Condés-Lara et al., 2009; Godínez-Chaparro et al., 2016).

4.5. Final considerations and clinical implications

Since MDH is an important integration site for peripheral nociceptive signals to the nervous system, it is worth considering the possible clinical relevance of our findings, particularly within the context of orofacial pain disorders and headaches (including migraines). This point of view, becomes increasingly important, considering that in human tissue, using competitive binding receptor autoradiography (to avoid bias detection with the $V_{1A}R$) for visualizing OTR with ^{125}I -ornithine vasotocin (a specific oxytocin radioligand) showed that oxytocin binding sites are present at spinal trigeminal nuclei (Freeman et al., 2017); most importantly $V_{1A}R$ was not detected (using SR-49059). Similar results were obtained in other Primates (*Macaca mulatta*, Freeman et al., 2014a; *Callicebus cupreus*, Freeman et al., 2014b).

Nevertheless, we must keep in mind one important limitation. Our electrophysiological experiments were performed in male rats and the oxytocin has been implicated in the sexual dimorphism observed in various processes (e.g. reproductive, learning, memory and social behavior) (Gimpl and Fahrenholz, 2001). Indeed, the density of oxytocin binding sites at central nervous system are linked to the gonadal steroids hormones (Tribollet et al., 1990; Kremarik et al., 1991). Certainly, using semiquantitative

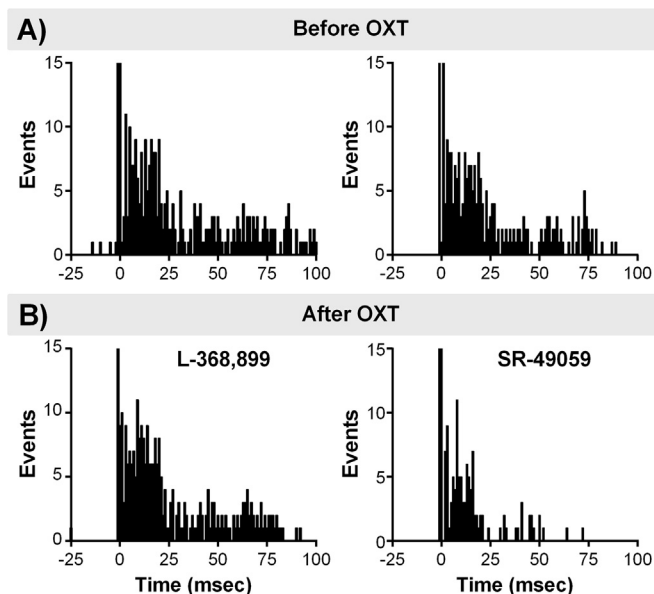


Fig. 5. Peri-stimulus time histograms obtained before (A; basal) and 40 min after (B) oxytocin (20 nmol) plus OTR (left) or $V_{1A}R$ (right) antagonists. Note that after pre-treatment with the selective OTR antagonist (20 nmol L-368,899) the oxytocin-induced antinociception was abolished whereas 20 nmol SR-49059 had no effect on the oxytocin effect.

autoradiography at spinal cord of adult rats Uhl-Bronner et al. (2005), showed that the oxytocin binding sites are denser in male than females rats. This finding correlate with a recent behavioral study (carrageenan-induced inflammation) showing that intrathecal injection of 0.125 nmol oxytocin in male rats elicited antinociception whereas to observe the same effect in female rats was necessary 1.25 nmol (Chow et al., 2016).

5. Conclusion

Using an electrophysiological and pharmacological approach, we show that oxytocin inhibit the peripheral electrically-induced neuronal firing of MDH nociceptive neurons by directly activating OTR. Taken together, our data show that oxytocin receptors at this level inhibits the peripheral nociceptive input arriving from the first branch of trigeminal nerve.

Authors contributions

All authors have read and approved the manuscript. EGB conducted the electrophysiological experiments. EGB and GML performed histological analysis and critically revised the manuscript. AGH and EGB performed data analysis. AGH and MCL designed experiments, interpreted data and wrote the manuscript. All authors read and approved the final version of the manuscript.

Funding

This work was supported by the Programa de Apoyo a Proyectos de Investigación e Innovación Tecnológica (PAPIIT-UNAM Mexico, grant No. IN200415 to MCL and IA203117 to AGH).

Conflicts of interest

The authors declare no conflict of interest. The sponsors had no further role in (1) study design, (2) collection, analysis and

interpretation of data, (3) the writing of the report and (4) the decision to submit the paper for publication.

Acknowledgements

The authors thank Jessica González Norris for proofreading the manuscript. EGB is a CONACyT-Mexico MSc Fellow.

References

- Abdallah, K., Artola, A., Monconduit, L., Dallel, R., Luccarini, P., 2013. Bilateral descending hypothalamic projections to the spinal trigeminal nucleus caudalis in rats. *PLoS One* 8, e73022.
- Akerman, S., Holland, P.R., Goadsby, P.J., 2011. Diencephalic and brainstem mechanisms in migraine. *Nat. Rev. Neurosci.* 12, 570–584.
- Alexander, S.P.H., Davenport, A.P., Kelly, E., Marrion, N., Peters, J.A., Benson, H.E., Faccenda, E., Pawson, A.J., Sharman, J.L., Southan, C., Davies, J.A., CGTP Collaborators, 2015. The Concise Guide to PHARMACOLOGY 2015/16: G protein-coupled receptors. *Br. J. Pharmacol.* 172, 5744–5869.
- Breton, J.D., Veinante, P., Uhl-Bronner, S., Vergnano, A.M., Freund-Mercier, M.J., Schlichter, R., Poisbeau, P., 2008. Oxytocin-induced antinociception in the spinal cord is mediated by a subpopulation of glutamatergic neurons in lamina I-II which amplify GABAergic inhibition. *Mol. Pain* 4, 19.
- Brito, G.A., Sachs, D., Cunha, F.Q., Vale, M.L., Lotufo, C., Ferreira, S.H., Ribeiro, R., 2006. Peripheral antinociceptive effect of pertussis toxin: activation of the arginine/NO/cGMP/PKG/ATP-sensitive K^+ channel pathway. *Eur. J. Neurosci.* 24, 1175–1181.
- Caldwell, J.D., Mason, G.A., Stanley, D.A., Jerdack, G., Hruby, V.J., Hill, P., Prange, A.J., Pedersen, C.A., 1987. Effects of nonapeptide antagonists on oxytocin- and arginine-vasopressin-induced analgesia in mice. *Regul. Pept.* 18, 233–241.
- Chow, L.H., Chen, Y.H., Wu, W.C., Chang, E.P., Huang, E.Y.K., 2016. Sex Difference in Oxytocin-induced anti-hyperalgesia at the spinal level in rats with intraplantar carrageenan-induced inflammation. *PLoS One* 11, 0162218.
- Condés-Lara, M., Martínez-Lorenzana, G., Rubio-Beltrán, E., Rodríguez-Jiménez, J., Rojas-Piloni, G., González-Hernández, A., 2015. Hypothalamic paraventricular nucleus stimulation enhances c-Fos expression in spinal and supraspinal structures related to pain modulation. *Neurosci. Res.* 98, 59–63.
- Condés-Lara, M., Rojas-Piloni, G., Martínez-Lorenzana, G., López-Hidalgo, M., Rodríguez-Jiménez, J., 2009. Hypothalamospinal oxytocinergic antinociception is mediated by GABAergic and opiate neurons that reduce A-delta and C fiber primary afferent excitation of spinal cord cells. *Brain Res.* 1247, 38–49.
- Condés-Lara, M., Rojas-Piloni, G., Martínez-Lorenzana, G., Rodríguez-Jiménez, J., López-Hidalgo, M., Freund-Mercier, M.J., 2006. Paraventricular hypothalamic influences on spinal nociceptive processing. *Brain Res.* 1081, 126–137.
- Condés-Lara, M., Zayas-Gonzalez, H., Manzano-García, A., Córdova-Quiroz, E., Granados-Mortera, J., García-Cuevas, M., Morales-Gómez, J., González-Hernández, A., 2016. Successful pain management with epidural oxytocin. *CNS Neurosci. Ther.* 22, 532–534.
- Eliava, M., Melchior, M., Knobloch-Bollmann, H.S., Wahis, J., da Silva Gouveia, M., Tang, Y., Ciobanu, A.C., del Rio, R.T., Roth, L.C., Althammer, F., Chavant, V., 2016. A new population of parvocellular oxytocin neurons controlling magnocellular neuron activity and inflammatory pain processing. *Neuron* 89, 1291–1304.
- Freeman, S.M., Inoue, K., Smith, A.L., Goodman, M.M., Young, L.J., 2014a. The neuroanatomical distribution of oxytocin receptor binding and mRNA in the male rhesus macaque (*Macaca mulatta*). *Psychoneuroendocrinol.* 45, 128–141.
- Freeman, S.M., Smith, A.L., Goodman, M.M., Bales, K.L., 2017. Selective localization of oxytocin receptors and vasopressin 1a receptors in the human brainstem. *Soc. Neurosci.* 12, 113–123.
- Freeman, S.M., Walum, H., Inoue, K., Smith, A.L., Goodman, M.M., Bales, K.L., Young, L.J., 2014b. Neuroanatomical distribution of oxytocin and vasopressin 1a receptors in the socially monogamous coppery titi monkey (*Callicebus cupreus*). *Neuroscience* 273, 12–23.
- Ge, Y., Lundeberg, T., Yu, L.C., 2002. Blockade effect of mu and kappa opioid antagonists on the anti-nociception induced by intra-periaqueductal grey injection of oxytocin in rats. *Brain Res.* 927, 204–207.
- Gimpl, G., Fahrenholz, F., 2001. The oxytocin receptor system: structure, function, and regulation. *Physiol. Rev.* 81, 629–683.
- Goadsby, P.J., Charbit, A.R., Andreou, A.P., Akerman, S., Holland, P.R., 2009. Neurobiology of migraine. *Neuroscience* 161, 327–341.
- Godínez-Chaparro, B., Martínez-Lorenzana, G., Rodríguez-Jiménez, J., Manzano-García, A., Rojas-Piloni, G., Condés-Lara, M., González-Hernández, A., 2016. The potential role of serotonergic mechanisms in the spinal oxytocin-induced antinociception. *Neuropeptides* 60, 51–60.
- Gong, L., Gao, F., Li, J., Yu, X., Ma, X., Zheng, W., Cui, S., Liu, K., Zhang, M., Kunze, W., Liu, C.Y., 2015. Oxytocin-induced membrane hyperpolarization in pain-sensitive dorsal root ganglia neurons mediated by Ca^{2+} /nNOS/NO/K ATP pathway. *Neuroscience* 289, 417–428.
- González-Hernández, A., Manzano-García, A., Martínez-Lorenzana, G., Tello-García, I.A., Carranza, M., Arámburo, C., Condés-Lara, M., 2017. Peripheral oxytocin receptors inhibit the nociceptive input signal to spinal dorsal horn wide-dynamic-range neurons. *Pain* 158, 2117–2128.
- González-Hernández, A., Rojas-Piloni, G., Condés-Lara, M., 2014. Oxytocin and

- analgesia: future trends. *Trends Pharmacol. Sci.* 35, 549–551.
- Gupta, J., Russell, R.J., Wayman, C.P., Hurlley, D., Jacson, V.M., 2008. Oxytocin-induced contractions within rat and rabbit ejaculatory tissues are mediated by vasopressin V1A receptors and not oxytocin receptors. *Br. J. Pharmacol.* 155, 118–126.
- Hobo, S., Hayashida, K.I., Eisenach, J.C., 2012. Oxytocin inhibits the membrane depolarization-induced increase in intracellular calcium in capsaicin sensitive sensory neurons: a peripheral mechanism of analgesic action. *Anesth. Analg.* 114, 442–449.
- Hu, J.W., 1990. Response properties of nociceptive and non-nociceptive neurons in the rat's trigeminal subnucleus caudalis (medullary dorsal horn) related to cutaneous and deep craniofacial afferent stimulation and modulation by diffuse noxious inhibitory controls. *Pain* 41, 331–345.
- Hu, J.W., Sun, K.Q., Vernon, H., Sessle, B.J., 2005. Craniofacial inputs to upper cervical dorsal horn: implications for somatosensory information processing. *Brain Res.* 1044, 93–106.
- Kaube, H., Hoskin, K.L., Goadsby, P.J., 1993. Inhibition by sumatriptan of central trigeminal neurones only after blood-brain barrier disruption. *Br. J. Pharmacol.* 109, 788–792.
- Kogelman, L.J.A., Christensen, R.E., Pedersen, S.H., Bertalan, M., Hansen, T.F., Jansen-Olesen, I., Olesen, J., 2017. Whole transcriptome expression of trigeminal ganglia compared to dorsal root ganglia in *Rattus Norvegicus*. *Neuroscience* 350, 169–179.
- Kremarik, P., Freund-Mercier, M.J., Stoeckel, M.E., 1991. Autoradiographic detection of oxytocin- and vasopressin-binding sites in various subnuclei of the bed nucleus of the stria terminalis in the rat. Effects of functional and experimental sexual steroid variations. *J. Neuroendocrinol.* 3, 689–698.
- Kubo, A., Shinoda, M., Katagiri, A., Takeda, M., Suzuki, T., Asaka, J., Yeomans, D.C., Iwata, K., 2017. Oxytocin alleviates orofacial mechanical hypersensitivity associated with infraorbital nerve injury through vasopressin-1A receptors of the rat trigeminal ganglia. *Pain* 158, 649–659.
- Lundeberg, T., Uvnäs-Moberg, K., Agren, G., Bruzelius, G., 1994. Anti-nociceptive effects of oxytocin in rats and mice. *Neurosci. Lett.* 170, 153–157.
- Mack, S.O., Wu, M., Kc, P., Haxhiu, M.A., 2007. Stimulation of the hypothalamic paraventricular nucleus modulates cardiorespiratory responses via oxytocinergic innervation of neurons in Pre-Bötzinger complex. *J. Appl. Physiol.* 102, 189–199.
- Malick, A., Strassman, R.M., Burstein, R., 2000. Trigemino-hypothalamic and reticulohypothalamic tract neurons in the upper cervical spinal cord and caudal medulla of the rat. *J. Neurophysiol.* 84, 2078–2112.
- Meng, I.D., Hu, J.W., Benetti, A.P., Bereiter, D.A., 1997. Encoding of corneal input in two distinct regions of the spinal trigeminal nucleus in the rat: cutaneous receptive field properties, responses to thermal and chemical stimulation, modulation by diffuse noxious inhibitory controls, and projections to the parabrachial area. *J. Neurophysiol.* 77, 43–56.
- Mørch, C.D., Hu, J.W., Arendt-Nielsen, L., Sessle, B.J., 2007. Convergence of cutaneous, musculoskeletal, dural and visceral afferents onto nociceptive neurons in the first cervical dorsal horn. *Eur. J. Neurosci.* 26, 142–154.
- Pettibone, D.J., Clineschmidt, B.V., Guidotti, M.T., Lis, E.V., Reiss, D.R., Woyden, C.J., Bock, M.G., Evans, B.E., Freidinger, R.M., Hobbs, D.W., Veber, D.F., 1993. L-368,899, a potent orally active oxytocin antagonist for potential use in preterm labor. *Drug Dev. Res.* 30, 129–142.
- Robert, C., Bourgeois, L., Arreto, C.D., Condes-Lara, M., Nosedá, R., Jay, T., Villanueva, L., 2013. Paraventricular hypothalamic regulation of trigeminovascular mechanisms involved in headaches. *J. Neurosci.* 33, 8827–8840.
- Rojas-Piloni, G., López-Hidalgo, M., Martínez-Lorenzana, G., Rodríguez-Jiménez, J., Condes-Lara, M., 2007. GABA-mediated oxytocinergic inhibition in dorsal horn neurons by hypothalamic paraventricular nucleus stimulation. *Brain Res.* 1137, 69–77.
- Serradeil-Le Gal, C., Wagnon, J., García, C., Lacour, C., Guiraudou, P., Christophe, B., Villanova, G., Nisato, D., Maffrand, J.P., Le Fur, G., 1993. Biochemical and pharmacological properties of SR 49059, a new, potent, nonpeptide antagonist of rat and human vasopressin V1a receptors. *J. Clin. Invest.* 92, 224–231.
- Shankland, W.E., 2001. The trigeminal nerve. Part II: the ophthalmic division. *Cranio* 19, 8–12.
- Shimizu, T., Suzuki, N., 2011. Biological sciences related to headache. In: Nappi, G., Moskowitz, M.A. (Eds.), *Handbook of Clinical Neurology*, vol. 97. Elsevier, Amsterdam, pp. 35–45.
- Schorscher-Petcu, A., Sotocinal, S., Ciura, S., Dupré, A., Ritchie, J., Sorge, R.E., Crawley, J.N., Hu, S.B., Nishimori, K., Young, L.J., Tribollet, E., 2010. Oxytocin-induced analgesia and scratching are mediated by the vasopressin-1A receptor in the mouse. *J. Neurosci.* 30, 8274–8284.
- Storer, R.J., Akerman, S., Goadsby, P.J., 2004. Calcitonin gene-related peptide (CGRP) modulates nociceptive trigeminovascular transmission in the cat. *Br. J. Pharmacol.* 142, 1171–1181.
- Swanson, L.W., McKellar, S., 1979. The distribution of oxytocin- and neurophysin-stained fibers in the spinal cord of the rat and monkey. *J. Comp. Neurol.* 188, 87–106.
- Thompson, K.L., Vincent, S.H., Miller, R.R., Colletti, A.E., Alvaro, R.F., Wallace, M.A., Feeney, W.P., Chiu, S.H.L., 1997. Pharmacokinetics and disposition of the oxytocin receptor antagonist L-368,899 in rats and dogs. *Drug Metab. Dispos.* 25, 1113–1118.
- Tribollet, E., Audigier, S., Dubois-Dauphin, M., Dreifuss, J.J., 1990. Gonadal steroids regulate oxytocin receptors but not vasopressin receptors in the brain of male and female rats. An autoradiographical study. *Brain Res.* 511, 129–140.
- Tsai, C.M., Chiang, C.Y., Yu, X.M., Sessle, B.J., 1999. Involvement of trigeminal subnucleus caudalis (medullary dorsal horn) in craniofacial nociceptive reflex activity. *Pain* 81, 115–128.
- Tzabazis, A., Kori, S., Mechanic, J., Miller, J., Pascual, C., Manering, N., Carson, D., Klukinov, M., Spierings, E., Jacobs, D., Cuellar, J., Frey 2nd, W.H., Hanson, L., Angst, M., Yeomans, D.C., 2017. Oxytocin and migraine headache. *Headache* 57, 64–75.
- Tzabazis, A., Mechanic, J., Miller, J., Pascual, C., Manering, N., Carson, D., Klukinov, M., Spierings, E., Jacobs, D., Cuellar, J., Frey 2nd, W.H., Hanson, L., Angst, M., Yeomans, D.C., 2016. Oxytocin receptor: expression in the trigeminal nociceptive system and potential role in the treatment of headache disorders. *Cephalalgia* 36, 943–950.
- Uhl-Bronner, S., Waltisperger, E., Martínez-Lorenzana, G., Condes-Lara, M.C., Freund-Mercier, M.J., 2005. Sexually dimorphic expression of oxytocin binding sites in forebrain and spinal cord of the rat. *Neuroscience* 135, 147–154.
- Wang, J.W., Lundeberg, T., Yu, L.C., 2003. Antinociceptive role of oxytocin in the nucleus raphe magnus of rats, an involvement of μ -opioid receptor. *Reg. Pept.* 115, 153–159.
- Wang, Y.L., Yuan, Y., Yang, J., Wang, C.H., Pan, Y.J., Lu, L., Wu, Y.Q., Wang, D.X., Lv, L.X., Li, R.R., Xue, L., 2013. The interaction between the oxytocin and pain modulation in headache patients. *Neuropeptides* 47, 93–97.
- Williams, P.D., Anderson, P.S., Ball, R.G., Bock, M.G., Carroll, L., Chiu, S.H., Clineschmidt, B.V., Culbertson, J.C., Erb, J.M., 1994. 1-((7, 7-Dimethyl-2 (S)-(2 (S)-amino-4-(methylsulfonyl) butyramido) bicyclo [2.2.1] heptan-1 (S)-yl) methyl) sulfonyl)-4-(2-methylphenyl) piperazine (L-368,899): an orally bioavailable, non-peptide oxytocin antagonist with potential utility for managing preterm labor. *J. Med. Chem.* 37, 565–571.
- Wilson, L.A., Wayman, C.P., Jackson, M., 2009. Neuropeptide modulation of a lumbar spinal reflex: potential implications for female sexual function. *J. Sex. Med.* 6, 947–957.